THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** physical change	Only form or state changes.
2	**(a)** melting of wax	Wax solidifies again.
3	**(a)** air and moisture	Both oxygen and water.
4	**(d)** chemical change	New substances appear.
5	**(d)** ripening of fruit	Cannot be undone.
6	**(a)** hydrated iron oxide	Reddish-brown flakes.
7	**(d)** physical change	Same substance, different state.
8	**(c)** burning of magnesium ribbon	New substance forms.
9	**(a)** painting	Coating blocks air and water.
10	**(d)** chemical change	Ash and gases form.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No new substance is formed (1); Melting of ice, tearing paper (1).	1+1
12	Presence of oxygen (1); Presence of moisture (1).	1+1
13	Fizzing (1); Carbon dioxide forms (1).	1+1
14	New substances are formed (1); Burning wood, rusting (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Melting: physical (1); Burning and rusting: chemical (1); Boiling: physical (1).	1+1+1
16	Mix them in a bottle (1); Gas formed (1); Pass through lime water (1).	1+1+1
17	Food gone bad smells (1); Milk turns sour (1); New substances (1).	1+1+1
	OR: Three test tubes with nails (1); Dry air, boiled water with oil, moist air (1); Only moist air tube rusts (1).	1+1+1
18	Natural: rain, seasons, day-night (1); Man-made: cooking, building, burning (1); Natural without human action (1).	1+1+1
	OR: Wax melts: physical (1); Wax vapour burns: chemical (1); Soot and gases formed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Physical: melting (1); Physical: tearing paper (1); Chemical: burning (1); Chemical: rusting (1); Chemical: souring of milk (1).	1+1+1+1+1
	OR: Colour change (1); Gas evolution (1); Heat and light (1); Precipitate (1); Smell change (1).	1+1+1+1+1
20	Iron + oxygen + water gives rust (1); Needs air and moisture (1); Paint (1); Galvanise (1); Alloy as stainless steel (1).	1+1+1+1+1
	OR: Melting of ice reversible (1); Folding paper reversible (1); Burning irreversible (1); Cooking irreversible (1); Depends on new substance (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** stretching a rubber band	Shape only.
2	**(d)** zinc	It protects iron.
3	**(d)** chemical change	Iron oxide forms.
4	**(c)** the moon's phases	Repeats at regular intervals.
5	**(b)** chromium and nickel	Chromium forms a protective layer.
6	**(c)** physical change	Sugar can be recovered.
7	**(c)** rusting of iron	Takes days.
8	**(c)** evaporation and cooling	Crystallisation.
9	**(a)** chemical change	New substances form.
10	**(d)** burning of a cracker	Takes seconds.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Painting (1); Galvanisation (1).	1+1
12	Copper is deposited (1); A new substance forms (1).	1+1
13	Reversible changes can be undone (1); Irreversible ones cannot (1).	1+1
14	Formation of crystals from a saturated solution (1); On cooling or evaporation (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Iron coated with zinc (1); Zinc is more reactive and corrodes first (1); Protects even if the layer is scratched (1).	1+1+1
16	Heat changes the substance (1); New substances are formed (1); Cannot be uncooked (1).	1+1+1
17	Substance remains the same (1); It can be recovered (1); No new product (1).	1+1+1
	OR: Make saturated solution (1); Filter and cool (1); Crystals form (1).	1+1+1
18	Burns with white light (1); White ash is formed (1); New substance magnesium oxide (1).	1+1+1
	OR: Food is broken down (1); New substances formed (1); Cannot be reversed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tube A: dry air (1); Tube B: boiled water and oil layer (1); Tube C: moist air (1); Observe after several days (1); Only C rusts (1).	1+1+1+1+1
	OR: Fizzing (1); Carbon dioxide released (1); Pass through lime water (1); It turns milky (1); A chemical change (1).	1+1+1+1+1
20	Heat water and add the salt (1); Stir till saturated (1); Filter (1); Cool slowly (1); Collect crystals (1).	1+1+1+1+1
	OR: Stainless steel resists rust (1); Zinc coating protects (1); Saves money (1); Saves resources (1); Painting also helps (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that every change of state is chemical; melting, freezing and boiling are **physical** changes.
- Calling dissolving of sugar or salt in water a chemical change; the substance can be recovered by evaporation.
- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
- Calling crystallisation a chemical change; crystals form without any new substance.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Changes Around Us: Physical and Chemical** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** crystallisation	Crystals form on cooling.
2	**(c)** physical change	Only shape changes.
3	**(d)** formation of rain	Occurs in nature.
4	**(d)** cooling the saturated solution	Crystallisation.
5	**(c)** chemical change	Food changes irreversibly.
6	**(a)** cooking of food	Done by humans.
7	**(d)** carbon dioxide	Fizzing.
8	**(b)** physical change	Water vapour is still water.
9	**(d)** chemical change	New substances formed.
10	**(d)** milky	CO_2 test.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Change in colour (1); Evolution of gas (1).	1+1
12	Cooking food (1); Souring of milk (1).	1+1
13	Wax melts: physical (1); Wax burns to new products: chemical (1).	1+1
14	No (1); Salt can be recovered by evaporation (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Colour changes (1); Taste changes (1); New substances form (1).	1+1+1
16	Phases of the Moon (1); Repeats at regular intervals (1); Also day and night (1).	1+1+1
17	Keep nails clean (1); Seal the tubes (1); Observe for several days (1).	1+1+1
	OR: Rusting is slow (1); Rust cannot become iron again easily (1); So irreversible (1).	1+1+1
18	Slow: rusting (1); Fast: burning of a cracker (1); Time differs (1).	1+1+1
	OR: Iron + oxygen + water (1); gives hydrated iron oxide (1); Called rust (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Wax melts and solidifies (1); Physical changes (1); Wax vapour burns (1); Produces carbon dioxide and water (1); Chemical change (1).	1+1+1+1+1
	OR: Rusting (1); Ripening (1); Growth of a plant (1); Burning cracker (1); Bursting balloon (1).	1+1+1+1+1
20	Cooking (1); Curdling (1); Rusting (1); Burning fuel (1); Photosynthesis (1).	1+1+1+1+1
	OR: Rain (1); Seasons (1); Cooking (1); Making a house (1); Making ice (1).	1+1+1+1+1

Mistakes to avoid:

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- Writing that rusting needs only air or only water; it needs **both** oxygen and moisture.
- Assuming all physical changes are reversible and all chemical changes irreversible without examples; ask whether a *new substance* forms.
- Writing that galvanisation uses copper or paint; the coating is of **zinc**.
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